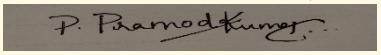
SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

1. CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1021	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
Student Name:	P. PRAMOD KUMAR	**Reg. No.:**	1925260245
Date:	29-10-26	**Duration:**	60 minutes

1.1 Code of Conduct

I P. PRAMOD KUMAR(1925260245) _ *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.* Signature of the Student: 

Annexure A – Scenario-Based Requirement Analysis

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functiona l Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete,	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
	accurate, and follows UML standards.			
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/con straints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	**Mark**
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25

Industry Context

Job markets are moving away from traditional four-year degrees toward continuous skills development and micro-credentials. Applicants may struggle to compile, share, and verify certificates from online courses, bootcamps, and workplace training. Employers face slow and expensive verification steps. A secure lifelong-learning registry can help individuals verify skills quickly.

Problem Statement

Design and develop a decentralized Lifelong Learning Ledger that allows educational institutions, employers, and online platforms to issue unalterable micro-credentials. It should manage skill badges, course completions, and professional certifications; allow learners to aggregate milestones into a digital portfolio; and let them share credentials with recruiters for rapid verification without depending on a central school registrar.

Objectives

• Build a decentralized platform to issue, claim, and organize digital skill credentials.

• Implement cryptographic validation layers to prevent document tampering or forgery.

• Create secure sharing controls so users manage who can view specific credentials.

• Integrate with HR hiring tools to automate skill verification during job applications.

• Support flexible micro-credential updates as technologies and job roles evolve over time.

Expected Outcomes

• Fast, low-cost verification of professional skills for hiring managers and recruiters.

• Better career mobility through a verified portfolio of lifelong skills.

• Reduced administrative costs for schools and training academies through automated issuing.

• A clearer, data-driven understanding of workforce skills and education trends.

2. Scenario Understanding and Problem Analysis

The project aims to create a trusted and portable record of learning achievements across institutions, online platforms, bootcamps, and workplace training. The main issue is enabling trustworthy verification while preserving learner control and limiting unnecessary disclosure.

2.1 Stakeholders

Stakeholder	Role / need
Learners / students	Claim, organize, and share credentials; control portfolio visibility.
Educational institutions	Issue verified course-completion credentials and revoke/replace erroneous records.
Online learning platforms	Issue badges and certificates using defined issuer rules.

Employers / recruiters	Verify authenticity, issuer, status, and relevant skill evidence.
System administrators	Manage configuration, security, monitoring, and recovery.
Credential standards / identity services	Support consistent credential formats and identity checks.

2.2 Main Challenges

• Fragmented records across providers.

• Slow manual verification.

• Forgery and tampering risk.

• Limited learner control over sharing.

• Inconsistent formats and integration needs.

• A valid signature does not independently prove skill quality.

3. Functional Requirements

ID	Requirement
FR-01	The system shall authenticate learners, issuers, recruiters, and administrators.
FR-02	The system shall allow approved issuers to create and digitally sign micro-credentials.
FR-03	The system shall assign each credential a unique identifier and record its proof/status on the ledger.
FR-04	The system shall allow learners to claim credentials and organize them in a digital portfolio.
FR-05	The system shall support secure sharing through user-approved links or verifiable presentations.
FR-06	The system shall let recruiters verify issuer identity, credential integrity, status, and relevant metadata.
FR-07	The system shall support issuer-authorized revocation or correction with an auditable status update.
FR-08	The system shall notify users of credential issuance, sharing, expiry, or status changes.
FR-09	The system shall maintain audit events for issuance, verification, sharing, and administrative actions.
FR-10	The system shall expose documented APIs for approved HR and learning-platform integrations.

Non-Functional Requirements

Quality	Requirement / acceptance focus
Security	Role-based access, strong authentication, signed credentials, encryption in transit and at rest.
Privacy	Store only necessary proof/metadata on-chain; keep personal data off-chain where appropriate.
Integrity	Credential signatures and hashes must make unauthorized changes detectable.

Availability	Verification service should remain usable during ordinary service interruptions; define recovery targets.
Performance	Set measurable targets for verification latency and concurrent requests during prototype testing.
Scalability	Support growing issuers, learners, credentials, and verification requests through modular services.
Usability	Learners can share a credential with clear consent and recruiters receive a readable result.
Interoperability	Use documented schemas and APIs to support multiple education and HR systems.
Auditability	Timestamped events and status changes should be reviewable by authorized auditors.
Maintainability	Modular design, versioned APIs, tests, and documented operational procedures.

4. Actors and Use-Case Modelling

Actor	Responsibilities
Learner	Claims issued credentials; organizes portfolio; grants/revokes sharing permission.
Credential Issuer	Verifies achievement and issues a signed credential.
Recruiter / Employer	Requests and checks credential evidence with learner permission.
Platform Administrator	Manages access, configuration, monitoring, and service health.
External HR / LMS	Exchanges credential and verification data through authorized APIs.
Auditor	Reviews access and credential lifecycle events.

4.1 Use-Case Catalogue

ID	Use case	Actor
UC-01	Register / authenticate	Learner, Issuer, Recruiter, Administrator
UC-02	Issue micro-credential	Credential Issuer
UC-03	Claim credential	Learner
UC-04	Manage digital portfolio	Learner
UC-05	Share selected credential	Learner
ID	**Use case**	**Actor**
UC-06	Verify credential	Recruiter / Employer
UC-07	Revoke or correct credential	Credential Issuer
UC-08	Integrate with HR / LMS	External HR / LMS
UC-09	Review audit logs	Auditor / Administrator

4.2 Use-Case Description: Verify a Credential

Primary actor: Recruiter / Employer. Supporting actors: Learner, Credential Issuer, Verification API. Precondition: the recruiter has a legitimate verification request and the learner has shared the credential or presentation.

Main flow: open shared credential; retrieve issuer and status proof; validate signature/hash; check expiry and revocation; display result; record timestamped event. If proof is invalid, issuer unknown, credential revoked, or status service unavailable, display a clear not-verified or unavailable result.

5. Proposed System Architecture

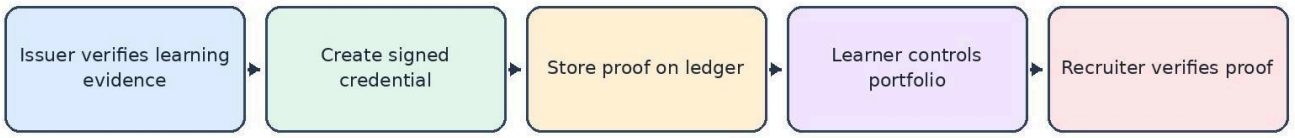
Separate the user interface, identity/access control, credential issuance/signing, ledger proof, off-chain storage, verification APIs, and audit/monitoring.

Proposed logical architecture



6. Real Workflow Diagram

Credential issuance and verification workflow



• Issuer verifies achievement and digitally signs credential.

• Ledger records proof/status reference.

• Learner claims and organizes credential in portfolio.

• Learner grants a recruiter permission to view selected credential.

• Recruiter verifies signature, issuer, status, and shared claims; event is logged.

7. Software Requirements Specification (SRS)

7.1 Scope

In scope: issuance, portfolio management, consent-based sharing, verification, revocation/status, audit logs, and HR/LMS integration. Out of scope: guaranteeing employment outcomes or certifying every course's quality.

7.2 Data Requirements

Entity	Attributes / purpose
Learner reference	Pseudonymous ID and account status

Issuer	Organization, issuer ID, public key/status
Credential	Unique ID, schema, achievement, dates, signature
Credential status	Active/revoked/expired state and timestamp
Share grant	Credential ID, recipient/scope, expiry, consent
Verification event	Request ID, result, timestamp
Audit event	Actor, action, object ID, timestamp

7.3 External Interfaces

• Responsive web portals for learners, issuers, and recruiters.

• Authenticated APIs for issuance, verification, status, and HR/LMS integration.

• Encrypted communication, validation, timeouts, and rate limits.

• Audit dashboard, alerts, key management, and recovery controls.

8. Assumptions and Constraints

Assumptions	Constraints
Issuers are onboarded and their signing keys are protected.	Personal information should not be placed permanently on a public ledger.
Credential definitions and issuer identities follow an agreed schema.	Credential status updates depend on issuer authorization and ledger/network availability.
Learners can access the internet and a supported identity/authentication method.	Interoperability requires agreement on credential schemas, identifiers, and APIs.
Recruiters have a legitimate purpose and receive only credentials the learner chooses to share.	Key loss, compromised issuer accounts, and incorrect source records require recovery procedures.
Ledger transactions and external integrations may incur fees or delays.	Prototype cost, development time, and available infrastructure limit initial scale.
Legal, privacy, retention, and accessibility requirements will be confirmed for the deployment region.	Deployment-specific legal and privacy obligations must be reviewed before launch.

9. Requirement Validation and Traceability

Validation item	Method / expected result
Completeness	Map each objective to functional/non-functional requirements and test cases.
Consistency	Check that sharing permissions, privacy rules, and verification outcomes do not conflict.
Unambiguity	Replace words such as 'instant' and 'secure' with measurable acceptance criteria.
Traceability	Link requirements to use cases, design components, and verification tests.
Integrity	Modify a test credential and confirm that signature/hash verification fails.

Access control	Test that unauthorized roles cannot issue, revoke, or view restricted credentials.
Recovery	Test issuer key rotation, revoked credentials, service outage, and audit recovery.
Usability	Ask representative learners and recruiters to complete a credential share/verify task.

10. Comparison: Traditional vs Proposed Ledger

Dimension	Traditional approach	Proposed ledger approach
Credential storage	Scattered PDFs, emails, and platform accounts	Unified learner portfolio with verifiable credential proofs
Verification	Manual calls and email checks	Cryptographic checks and status lookup
Tamper detection	Difficult to detect altered documents	Signature/hash validation reveals unauthorized changes
Learner control	Sharing often requires sending full files repeatedly	Permission-based sharing of selected credentials
Audit trail	Records split across organizations	Timestamped issuance, sharing, and status events
Interoperability	Different formats and portals	Documented schema and API integration (requires adoption)
Privacy	Copies may spread without clear tracking	Data minimization and consent controls; design must avoid exposing personal data

11. Risks and Mitigations

• Incorrect source credential: issuer checks and correction/revocation procedures.

• Compromised issuer key: protected keys, rotation, and incident response.

• Privacy exposure: data minimization, consent, and access control.

• Ledger/API outage: monitoring, retries, and recovery.

• Format fragmentation: schema governance and conformance tests.

• Avoid overstated trust: cryptographic verification proves integrity and issuer control, not full skill quality.

12. Implementation and Testing Plan

• Phase 1: define schemas, roles, consent, and acceptance criteria.

• Phase 2: build issuer portal, portfolio, signing service, and ledger proof recording.

• Phase 3: implement verification, status/revocation, and HR/LMS APIs.

• Phase 4: test tampering, access control, revocation, performance, usability, and recovery.

• Acceptance: tampered proof fails; revoked credential is flagged; unauthorized access is denied; valid sharedcredential verifies.

13. Conclusion

The proposed Lifelong Learning Ledger addresses fragmented credential records and slow manual verification using digitally signed credentials, tamper-evident ledger proofs, learner-controlled sharing, and verification APIs. It can

improve verification speed, career mobility, and administrative efficiency. Deployment requires trusted issuers, interoperable standards, privacy-preserving design, key protection, clear revocation, and measurable testing. Decentralization alone does not guarantee privacy, correctness, or availability.

14. References

- W3C, Verifiable Credentials Data Model v2.0, https://www.w3.org/TR/vc-data-model-2.0/
- W3C, Decentralized Identifiers (DIDs) v1.0, https://www.w3.org/TR/did-core/
- European Commission, European Digital Credentials for Learning, https://europa.eu/europass/digital-credentials/
- Open Badges Specification, 1EdTech Consortium, https://www.1edtech.org/standards/open-badges
- Source material supplied by the user: screenshots describing the industry context, problem statement, objectives,and expected outcomes.